

Internet of Things Integrated Smart Specially Abled Assistance System

Vaishnavi Namdeo	Adarsh Jha	Ishita Rastogi	Mohd. Amaan	Sachin Sharma
Department of Computer Science and Engineering Graphic Era Deemed to be University Dehradun, India	Department of Computer Science and Engineering Graphic Era Deemed to be University Dehradun, India	Department of Computer Science and Engineering Graphic Era Deemed to be University Dehradun, India	Department of Computer Science and Engineering Graphic Era Deemed to be University Dehradun, India	Department of Computer Science and Engineering Graphic Era Deemed to be University Dehradun, India
vaishnavinamdeo.21022855@gmail.com	jhaadarsh195@gmail.com	ishitarastogi.21022057@gmail.com	mohdamaanonline000@gmail.com	sxsharma88@gmail.com

Abstract— A recent report from March 2023 indicates that approximately 1.3 billion individuals worldwide have been identified as having disabilities or special needs, but a significant number of these children do not attend regular schools. These students often encounter discrimination, bullying, and a lack of equal opportunities in both public spaces and the educational system. While specialized institutions can provide some support, transitioning to the real world can be difficult for students as they are unfamiliar with environments that differ from their own. To address this issue, governments are increasingly promoting inclusive education, which offers a diverse learning atmosphere and enables students with disabilities to interact with those without disabilities. This paper proposes an assistance system that utilizes a web-application and wearable device, such as a wristband, equipped with integrated sensors and a camera. The system is designed to monitor the health of students with special needs and provide regular updates to their guardians. When trigger is pressed, a video recording is initiated and a call is made to the authority or guardian. The video is then saved to Google Drive, and the link to the video is sent via SMS. Through such measures, inclusive education can be enhanced to ensure the well-being and success of specially abled students.

Keywords—Internet of Things (IoT), Smart Monitoring, Alert System, ThingSpeak, ESP-32

I. INTRODUCTION

National Education Policy, 2020 guarantees 'equitable and comprehensive instruction to all'. It is majorly made to extend support of the children with inability in typical educate or schools. Approximately Eighty percent of Indian populace lives in country zones without arrangement for extraordinary schools [1]. In many developing countries of Asia, these children are often excluded from society and thus have very less participation in working sector or in directly contributing to economy. In the early 1990s, international studies on bullying and students with disabilities began. Bullying is up to 63 percent more likely to affect children with disabilities or learning difficulties than their peers [2]. Despite so many efforts from the government, the children with special needs face many challenges and their parent or guardian are often scared to send them to school. These students are either home-schooled or sent to specially abled institutions that leave them with very less chances of getting higher education or a job, forcing them to become a liability. Nowadays, due to disturbance in innovation parcel of modern things like – work automation, IoT, AI, machine learning AR, VR, etc. are changing the work environment, particularly for individuals with inabilities and requesting more nimbleness

in workforce to receive speedier (changes in ways of working and culture) and make comprehensive and available organizations [3]. Without any assistance it is hard for children with special needs to take part in normal jobs and education settings. Hence, we realized that an assistance system that can help the guardian in monitoring and tracking activities and health condition of their ward is needed. Sensors are devices that detect and respond to changes in the environment and convert them into measurable signals. IoT devices used:

- **Arduino Uno:** It is an open-source micro-controller board based on Atmega328, and highlights advanced and analog input/output pins. The Arduino Uno incorporates a add up to of 14 advanced input/output pins, of which 6 can be utilized as PWM (Pulse Width Modulation) yields. It moreover has 6 analog input pins, which can be utilized to examined analog sensors or input voltages. This controller has everything required to back the microcontroller by basically interfacing it to a computer with a USB cable or control it with an AC-to-DC connector [10].
- **Wifi-Module:** It is device that is integrated into the device to connect to wireless local area networks (WLANs) and communicate with other devices and the internet using WiFi technology. We have used ESP8266, which is most commonly used and is low-cost Wi-Fi microchip with full TCP/IP stack and microcontroller capability [9].

Some of the sensors used in our system are:

- **Temperature (DHT11):** To measure temperature of the child.
- **Heartbeat (KY-039):** To measure heart rate of the child.
- **Camera (ESP-32):** To record video.

II. LITERATURE SURVEY

According to “AI and IoT Enabled Technologies for Monitoring the Right to Health of Disabled People” [4] is also made, the focus is on how technology powered by AI can be useful to enhance the state of disabled people. So, they have provided some technological inferences that may help in supporting the People with Disabilities (PWD). But in this research, they only presented some theoretical ideas which may be implemented in the future. On the other hand, we are presenting theoretical ideas already with the proof of implementing it. During this research, we have also discovered a topic “Improving Cognitive Visual-Motor Abilities in Individuals with Down Syndrome” [5] which concentrates on a specific group of people unfortunately

diagnosed with Down Syndrome. Down Syndrome is a disability which occurs in a child because of an extra chromosome in the cell which leads to forgetting many things, unable to learn new things quickly, can have strong visual short-term memory and may have more health-related problems. In this, they expressed that the CVMA (Cognitive Visual Motor Ability) can be progressed in individuals by utilizing works out with a gestural interaction utilizing the KINECT sensor. Ready to take one more case on a term paper made on “Sensor Innovations for Caring Individuals with Disabilities” [6]. This one or maybe concentrates on the utilize of different sensors as a portion of innovation which can be utilized for individuals with incapacities. It points to show later inquires about and improvements on sensors advances which centers on diverse arrangements which can be conceivably beneficial on more precise perusing. It can be made for more sophistic observing and caution frameworks for uncommonly abled individuals. Amid this look, we have moreover found an application based on “IFSA: an coordinates system for creating IoT connected versatile applications for uncommonly abled people” [7]. It is so troublesome to utilize versatile applications or shrewd and complex innovations particularly for uncommonly abled individuals these days. This term paper mainly focuses on how to plan and give a system for creating a versatile application utilizing IoT to supply secure and all-in-one administrations to the outwardly disabled individuals. Think about told us different remote implanted frameworks utilizing portable phones can be utilized. One more brilliant illustration accessible can be on “iChat: Interactive Eyes for Specially Challenged People Using OpenCV Python” [8], which is primarily centered on the invention and clarification of eye following framework. This eye tracking framework can be utilized for the individuals who have no physical capacity but eyes. Their fundamental point is they can chat and sort without the utilize of hands or fingers which could be a boon or favoring for extraordinarily abled individuals or have misplaced the physical capacity or have loss of motion .

III. PROPOSED FRAMEWORK

The system comprises of a wristband or pendant-like device that features multiple sensors and is connected to a web application. To access the system, the guardian or authorized user must login to the web application and authenticate themselves. If authorized, they will be directed to the main page. Fig. 1 shows step-by-step execution of our framework. *Working of the proposed model:* The following steps outline the implementation of our proposed framework. The user or authority should log in to the web-application and proceed to the home page. The home page consists of different sections that the user can access. The first section, titled "Student/Ward Data," contains information about the student and their guardian. It includes the authority's phone number for emergency contact, as well as additional details such as the student's school or college, address, and roll number. It also provides information about the student's hospital and regular doctor, particularly for those who are specially abled. The second section, titled "Medical History," provides further details about the student's disability and any relevant government welfare or scholarship schemes that may be available to them. The most significant section, titled "Proctor," includes various subsections for monitoring the student's health remotely. It allows the guardian to track the student's temperature, heartbeat, and other vital signs. The "Proctor" section also features a "video" subsection that

includes a Google Drive link to a recorded video. In the event of an emergency or accident, the student can press the button integrated into their wristband or pendant, which activates the video recording and initiates a call to the authority. Additionally, an SMS message containing the Google Drive link to the recorded video is sent to the authority as shown in Fig. 2.

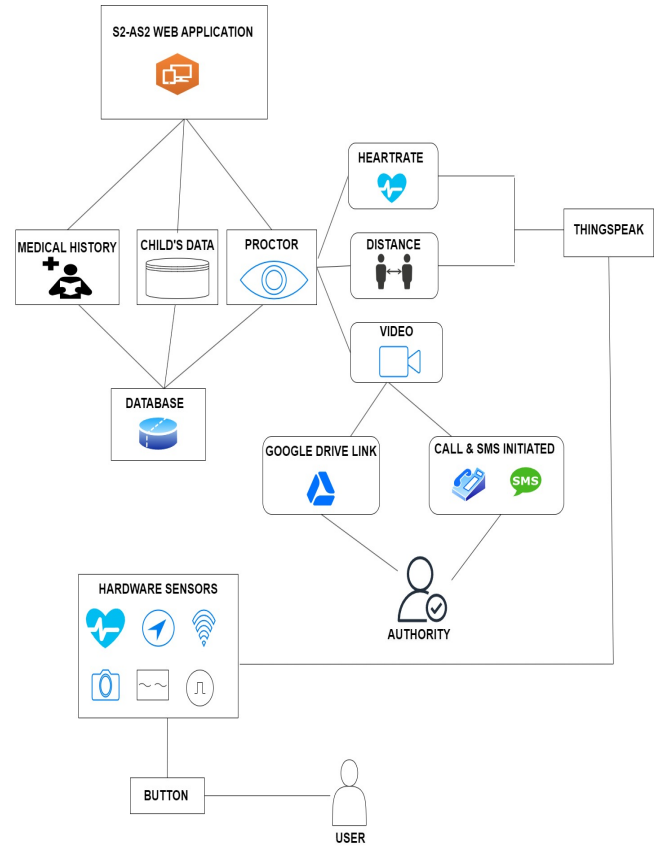


Fig. 1 Proposed framework.

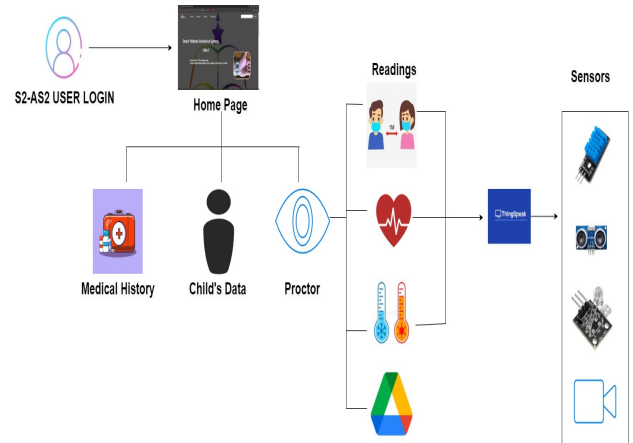


Fig. 2 Features of proposed framework.

Features: The device is lightweight and easy to wear, making it accessible to individuals with special needs and enhancing their sense of safety and security. Our web-based platform displays live sensor readings of a child's temperature and heartbeat [11]. The integrated ultrasonic sensor measures distance and sends an instant warning if someone comes too close to the student or poses a threat to their safety [12]. With this feature, authorities can monitor the well-being of children at any time and from any location.

Our application provides real-time notifications and automatic call and SMS alerts to authorities. Fig. 2 explores the features of proposed framework. The architecture of our proposed model can be broadly categorized into three primary components. The first component consists of the web frontend, which is accessible and user-friendly, enabling users to interact with the system and display the content provided. The second component is the integration of the IoT, which comprises all the sensors necessary for the proper functioning of the system. The third and final component is the backend, which houses the practical implementation and operations of the call and SMS alert system. Figure 4 graphically represents the architecture of our assistance system.

A. Frontend

The web-application is developed using Flask framework in Python along with web development tools like HTML and CSS to make it more user interactive. The home page looks like Fig. 3.



Fig. 3 Screenshot of front page.

Below the home page, we scroll down and find the following sections named 'Medical History' to display the disability of the child and related scholarship schemes for him/her. The second section is 'Child's Data' that consist of the student's name and guardian details. It also contains the student's school or college name, roll number and more. Fig. 4 shows the sections on app and Fig. 5, Fig. 6 and Fig. 7 display the subsections of these pages respectively.

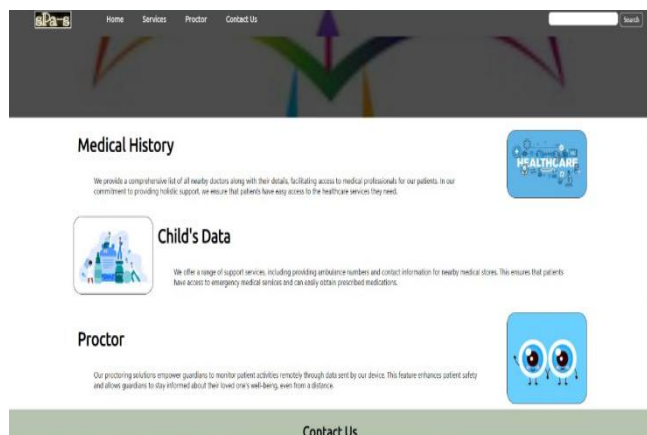


Fig. 4 Screenshot of front page.

Stella Rose

Add New Entry

Guardian's Name	Guardian's Phone	Guardian's Occupation	Guardian's Address	Hospital	Regular Doctor
John Doe	123-456-7890	Business	undefined	SpecialCare Hospital	Dr. Smith
Jane Smith	456-789-0123	Teacher	123 Royal Rd	Special Care Hospital	Dr. Smith

Fig. 5 Screenshot of sub section.

Guardian's Name:

Guardian's Phone:

Guardian's Occupation:

Guardian's Address:

Hospital:

Regular Doctor:

Submit

Guardian's Name	Guardian's Phone	Guardian's Occupation	Hospital	Regular Doctor
John Doe	123-456-7890	Bu	SpecialCare Hospital	Dr. Smith
Jane Smith	456-789-0123	Te	Special Care Hospital	Dr. Smith

Fig. 6 Screenshot of sub section.

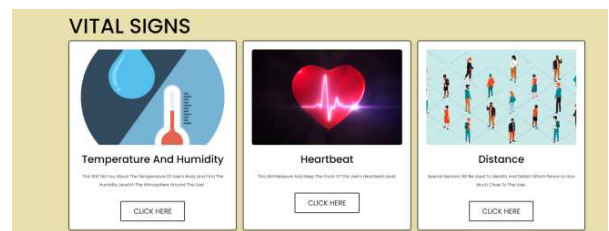


Fig. 7 Screenshot of sub section.

At the end of the app, we have a contact section for the user to send us their feedback or concern.

B. IoT Integration

In our model, we have incorporated various sensors to monitor the vital signs of a child. The sensors utilized include:

- **Push Button:** A push button is a type of switch that is actuated by pressing it. Once activated, it completes the electronic circuit, thereby sending a signal to the connected device or system, indicating that the button has been pressed [13]. There are various types of push button sensors available, including momentary and latching switches.
- **Temperature and Humidity Sensor (DHT11):** The DHT11 sensor contains a humidity sensing component and a thermistor to measure humidity and temperature, respectively. The data is transmitted in the form of 40-bit data packets [14]. It can measure temperature ranging

from 0°C to 50°C with an accuracy of $\pm 2^\circ\text{C}$ and humidity from 20% to 90% with an accuracy of $\pm 5\%$. The pin diagram for setting up this sensor.

- Ultrasonic Sensor:** The ultrasonic sensor measures the distance between the sensor (the person wearing it) and an object by calculating the time taken for ultrasonic waves to travel to the object and back [15]. It's primary purpose is to track any suspicious or irregular activity. If another person comes too close or attempts to harm the student/child, a warning will be instantly sent to the authority. The distance is calculated using the formula: $\text{distance} = (\text{time taken by ultrasonic waves to return} \times \text{speed of sound}) / 2$. Fig. 8 shows the pin diagrams and Fig. 9 shows the circuit diagram.

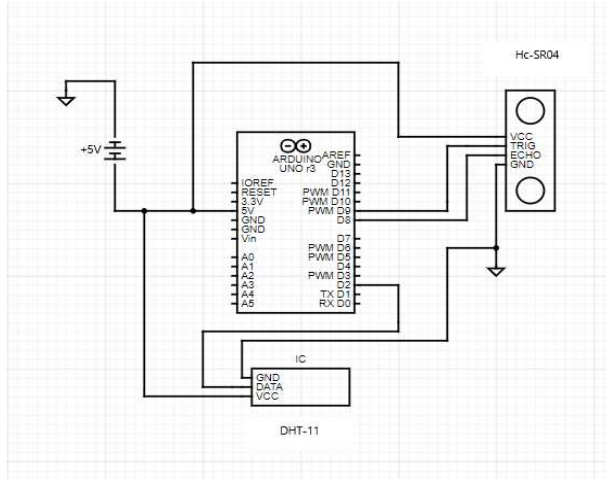


Fig. 8 Screenshot of DHT 11 and ultrasonic sensor's pin diagram.

Heartbeat or Pulse Sensor (KY-049): This sensor uses an analog output signal to detect heart rate. It employs an infrared LED and a photo detector to detect changes in blood volume. The accuracy and sensitivity may vary based on light conditions.

ESP32 Camera: The ESP32-CAM board includes an OV2640 camera module that can capture JPEG images or stream video and has an integrated wi-fi module that enables it to connect to other devices for data transmission. It also features a microSD card slot to store captured images and videos. The GPIO pins can be utilized for additional functionality, such as sensor integration [Fig. 10].

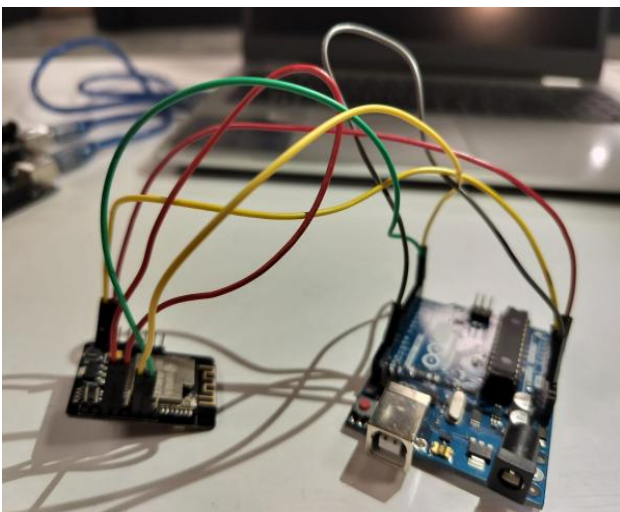


Fig. 9 Circuit diagram.

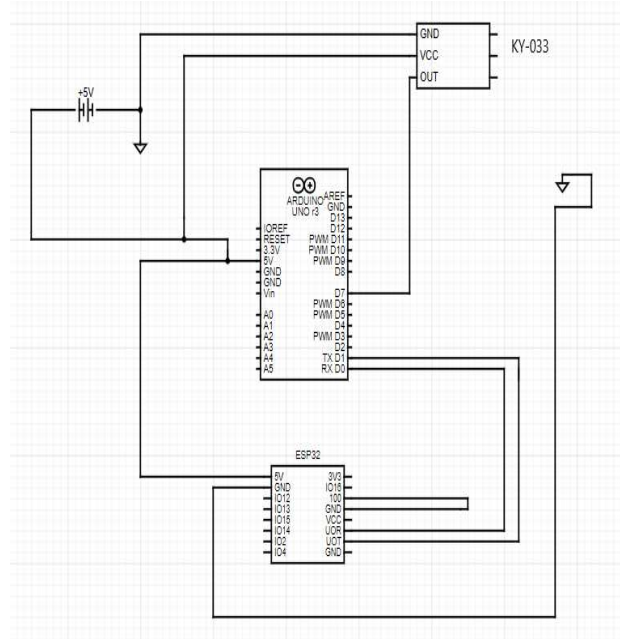


Fig. 10 Screenshot of sensor's pin diagram.

C. Backend

We have utilized Python libraries such as OpenCV and Twilio to capture video, save it to a file, initiate a voice call, and send an SMS notification when the recording commences.

Libraries used:

OpenCV: Usually a broadly utilized open-source library for computer vision and machine learning. It gives a extend of capacities for image and video handling. We utilized it to capture video from the webcam, show it, and store it in a record.

Os: This module enables interaction with the operating system. It allows Python code or scripts to perform various tasks that are dependent on the operating system. In our model, we employed it to check and delete files.

Twilio.rest: This module offers a Python API for interacting with Twilio services, such as initiating voice calls and sending SMS messages. It is a cloud-based communication platform that enables us to make or receive voice or phone calls and send or receive SMS messages.

Explanation or working of code:

- The script continuously captures video until the 'x' key is pressed.
- Upon activation of the recording, it sends an SMS notification and initiates a voice call.
- Lastly, it releases the resources and concludes the script.

IV. RESULT

The outcomes of our framework may be categorized into two segments, as delineated in the implementation section. The first comprises the distinct readings from each sensor that the user or supervisory authority can access via their app. These are the real-time measurements obtained from the student, which are transmitted from the wearable device, such as a band or pendant, to the ThingSpeak platform. This platform converts the data into a human-readable format, making it view-able by the guardian from any location and at any time. Fig. 11 shows the Temperature and humidity reading of ThingSpeak that is connected to the app.

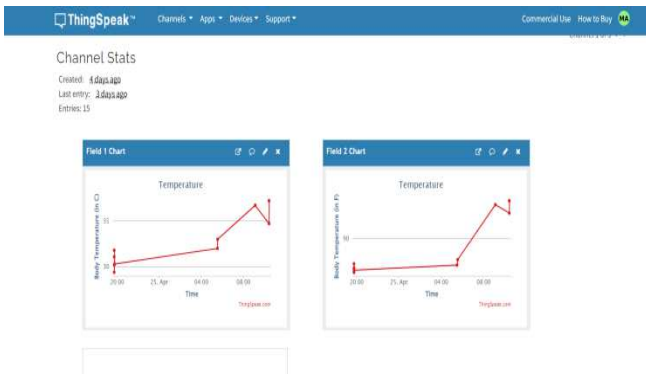


Fig. 11 Temperature and humidity reading of the specially abled child.

Fig. 12 shows the heartbeat reading of the specially abled child.

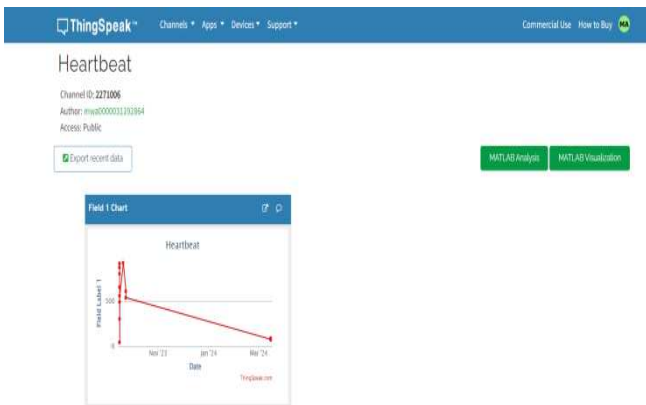


Fig. 12 Heartbeat reading of the specially abled child.

Fig. 13 shows the ultrasonic measurement that is distance and checks for any suspicious activity.

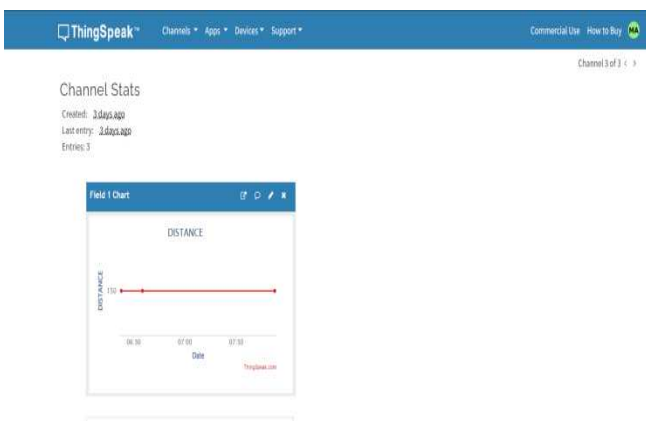


Fig. 13 Ultrasonic measurements.

The second scenario occurs when a child with special needs feels endangered or threatened. In such cases, the child may press the trigger or button. Upon pressing the button, the video recording function is activated, and a call is initiated to the designated authority, as illustrated in the aforementioned figure. The recorded video is saved to the Google drive, and a text message containing a link to the drive is also sent to the authority, allowing them to view the events that transpired involving their child. Fig. 14 shows the call that is initiated after the trigger is pressed. Fig. 15 shows the video

recorded being saved on Google drive. Fig. 16 shows the SMS received after the call, containing the Google Drive link.

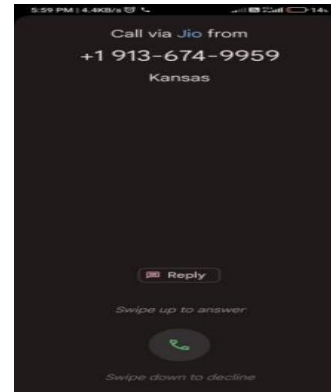


Fig. 14 Voice Call initiation.

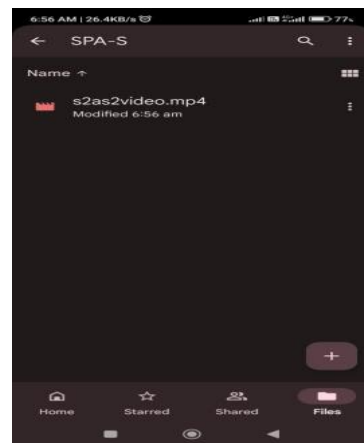


Fig. 15 Video Call initiation.

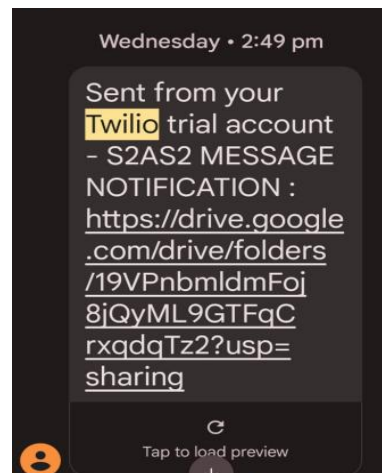


Fig. 16 SMS initiation.

V. FUTURE PERSPECTIVE

The future viewpoint of an IoT coordinates keen extraordinarily abled help framework is promising, with potential for critical headways in upgrading the quality of life for people with incapacities. Here's a see at a few angles of its future:

- Enhanced Accessibility: IoT innovations can make situations more available for individuals with incapacities. Savvy sensors implanted in different gadgets and situations can distinguish the nearness and needs of differently-abled people, altering settings

consequently to oblige them. For occurrence, keen homes can adjust lighting, temperature, and furniture courses of action based on the occupant's prerequisites.

- **Personalized Assistance:** Future frameworks can offer personalized help custom-made to person needs. By coordination machine learning calculations, these frameworks can learn from client intuitive and inclinations, giving more precise and productive back over time. This might incorporate updates for medicine, help with day by day errands, and indeed foreseeing and avoiding potential risks.
- **Mobility and Navigation:** IoT can revolutionize portability and route help for people with versatility impedances. Smart portability helps, such as powered wheelchairs and exoskeletons, can be prepared with IoT sensors and network highlights to explore situations independently, dodging deterrents and guaranteeing security.
- **Health Monitoring and Crisis Response:** IoT gadgets can continuously screen the wellbeing status of specially-abled people, collecting real-time information on imperative signs, pharmaceutical adherence, and action levels. In case of crises, such as falls or sudden wellbeing issues, these frameworks can consequently trigger alarms to caregivers or crisis administrations, giving opportune help.
- **Integration with Assistive Advances:** Future frameworks will likely coordinated with a wide extend of assistive advances, counting prosthetics, communication gadgets, and tactile helps. By interfacing these gadgets to the IoT biological system, clients can advantage from consistent communication and interoperability, improving their generally encounter and freedom.

VI. CONCLUSION

The paper presents a band or pendant model aimed at enhancing the safety and security of individuals with disabilities in educational institutions, public spaces, and other settings. The device allows authorities to remotely monitor the health and activities of the wearer. In case of an emergency, the wearer can trigger an instant call and SMS alert. However, the functionality of the model relies entirely on Wi-Fi connectivity, which may pose challenges in rural areas. To address this limitation, a future enhancement could involve incorporating a SIM card and an upgraded module to eliminate the dependency on Wi-Fi. Furthermore, additional sensors such as a Blood Pressure Controller for blood pressure monitoring, an ECG sensor for individuals with heart conditions, and GPS or GMS for location tracking and inclusion in the SMS alert could be integrated. A secondary or alternative phone number could also be added to ensure that the call is routed to another contact if the first one is unreachable. Additionally, the paper suggests the inclusion of a chat assistance feature on the website and the use of machine learning algorithms to determine whether a person with disabilities is in distress. In conclusion, the proposed device has the potential to significantly benefit individuals with special needs by providing them with increased safety and peace of mind in their daily lives.

REFERENCES

- [1] Singh, J. D. "Inclusive education in India—concept, need and challenges." S. No. Paper Title Author Name Page No 97 (2016).
- [2] Iqbal, Faiza, M. S. Senin, Mohd Norazmi Bin Nordin, and Muhammad Hasyim. "A qualitative study: Impact of bullying on children with special needs." *Linguistica Antverpiensia* 2 (2021): 1639-1643.
- [3] Garg, Shalini, and Shipra Sharma. "Impact of artificial intelligence in special need education to promote inclusive pedagogy." *International Journal of Information and Education Technology* 10, no. 7 (2020): 523-527.
- [4] Mehta, Kapil, Vandana Mohindru Sood, Chamkaur Singh, and Pratham Chabra. "Machine learning based intelligent system for safeguarding specially abled people." In 2022 7th international conference on communication and electronics systems (ICCES), pp. 1199-1206. IEEE, 2022.
- [5] Torres-Carrión, Pablo V., Carina S. González-González, Pedro A. Toledo-Delgado, Vanesa Muñoz-Cruz, Rosa Gil-Iranzo, Nuria Reyes-Alonso, and Selene Hernández-Morales. "Improving cognitive visual-motor abilities in individuals with down syndrome." *Sensors* 19, no. 18 (2019): 3984.
- [6] García-Peñalvo, Francisco José, and Manuel Franco-Martín. "Sensor technologies for caring people with disabilities." *Sensors* 19, no. 22 (2019): 4914.
- [7] Kaur, Sukhpreet, and Kanwalvir Singh Dhindsa. "IFSA: an integrated framework for developing IoT linked mobile applications for specially abled people." *Wireless Networks* 28, no. 4 (2022): 1375-1388.
- [8] Kathpal, Kashish, Shweta Negi, and Shilpi Sharma. "iChat: interactive eyes for specially challenged people using OpenCV Python." In 2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions)(ICRITO), pp. 1-5. IEEE, 2021.
- [9] Baig, Mirza Mansoor, and Hamid Gholamhosseini. "Smart health monitoring systems: an overview of design and modeling." *Journal of medical systems* 37 (2013): 1-14.
- [10] Kumar, S. Pradeep, Vemuri Richard Ranjan Samson, U. Bharath Sai, PLSD Malleswara Rao, and K. Kedar Eswar. "Smart health monitoring system of patient through IoT." In 2017 international conference on I-SMAC (IoT in social, mobile, analytics and cloud)(I-SMAC), pp. 551-556. IEEE, 2017.
- [11] Rajpoot, Navneet Kumar, Prabhdeep Singh, and Bhaskar Pant. "Cloud-IoT Based Smart Healthcare Framework: Application and Trends." In Proceedings of the 4th International Conference on Information Management & Machine Intelligence, pp. 1-6. 2022.
- [12] Sachin Sharma, Aanika Gupta, and Ranu Tyagi. "Artificial Intelligence, Cyber Security and Vedic Scripture Integrated Sustainable Global Youth Leadership Development." In 2024 IEEE International Conference on Communication, Computing and Signal Processing (IICCCS). IEEE, 2024.
- [13] Jiya Sharma, and Sachin Sharma. "Sustainable Solar Powered Bluetooth Enabled Smart Home Automation System Using Internet-of-Things." In 2024 IEEE International Conference on Communication, Computing and Signal Processing (IICCCS). IEEE, 2024.
- [14] Chetan Pandey, Sachin Sharma, and Shuchi Bhadula. "Artificial Intelligence Technique in Next Generation Healthcare Systems." In 2024 IEEE International Conference on Communication, Computing and Signal Processing (IICCCS). IEEE, 2024.
- [15] Stuti Bhatt, Sachin Sharma, and Shuchi Bhadula. "NeuroAI: Emerging Artificial Intelligence and Image Processing Techniques in Neuroscience for Enhanced Medical Diagnosis of Brain Tumor." In 2024 First International Conference on Pioneering Developments in Computer Science & Digital Technologies (IC2SDT), pp. 204-209. IEEE, 2024.